

Ishan Saxena

Software Engineer | Backend Developer

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Technical Skills

Languages	Java, C++, TypeScript, JavaScript (ES6+), Python
Backend Frameworks	Spring Boot, Spring MVC, Spring Data JPA, Hibernate, NestJS, Node.js
Databases & Caching	PostgreSQL, MongoDB, Redis
Cloud & DevOps	AWS (EC2, S3), Docker, CI/CD, Git
Architecture & APIs	System Design, Microservices, Distributed Systems, RESTful APIs
Tools & Security	JWT, OAuth 2.0, Postman, IntelliJ IDEA, VS Code, Figma

Experience

CirclePe

May 2025 – June 2025

Web Development Intern

- Built a blog admin panel for **15+ content types**, integrated **TipTap**, and shipped an SEO-ready **Area Converter tool** for production use.
- Programmed deployment-ready **data ingestion pipelines** processing **1,000+ data points**, producing city sitemaps and ensuring dependable downstream **data workflows**.
- Prototyped **UX/UI flows** in **Figma** and **React**, improving navigation across devices.
- Improved application performance and reliability using **React.js**, **Next.js**, and **Python**.

Projects

Scalable OTT Streaming Backend | System Design, PostgreSQL, Redis, AWS

[GitHub](#)

- Architected a distributed OTT backend supporting secure uploads of **multi-GB media assets**, creator-viewer workflows, JWT authentication, and **10+ independent backend services**.
- Engineered an asynchronous media-processing workflow using **FFmpeg**, **BullMQ**, and **Redis** to convert MP4 uploads into adaptive HLS streams with hundreds of video segments for efficient playback..
- Configured cloud-native delivery infrastructure using **AWS S3** and **CloudFront signed URLs**, enabling CDN-accelerated streaming, protected asset access, and low-latency media distribution.

BodhGanga — Educational E-Commerce Platform | Spring Boot, AWS, Docker

[Live App](#) | [GitHub](#)

- Modeled a distributed **Spring Boot REST API** containerized with **Docker** and deployed on **AWS EC2**, featuring **Razorpay** integration and a custom district-based content-unlock system.
- Enforced strict **Spring Security RBAC** alongside a secure **JWT + OTP (MSG91)** authentication processing layer to isolate user data and protect restricted educational resources.
- Formulated an automated **AWS S3** media orchestration and established a **GitHub Actions CI/CD** workflow, cutting deployment effort by **~60%** while maintaining over **99% operational uptime**.

SyncCanvas — Real-Time Collaborative Synchronization Engine | Spring Boot, WebSockets, Redis

[GitHub](#)

- Orchestrated a production-grade real-time synchronization engine using **Java 21**, **Spring Boot**, **WebSockets**, and **PostgreSQL**, following **Clean Architecture**, SOLID principles, and a modular layered architecture.
- Built secure REST APIs with **Spring Security** and **JWT authentication**, implementing board management, room lifecycle, DTO-based request/response models, validation, global exception handling, and persistent data storage.
- Streamlined low-latency multi-user collaboration through an authoritative server model, enabling room-based synchronization, shared state management, event ordering, presence tracking, and a scalable architecture designed for horizontal scaling using **Redis Pub/Sub**.

Education

B.Tech: Computer Science and Engineering

Sept 2023 – Present

VIT Bhopal University (CGPA: 8.8)

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, System Design, Web Technologies, Software Engineering

Certifications

- Google IT Support Professional Certificate**, Google – *IT Support, OS, Networking*
- McKinsey Forward Program**, McKinsey & Co. – *Business Communication, Leadership*
- C++ Programming Masterclass**, Udemy – *Advanced OOP, Memory Management*